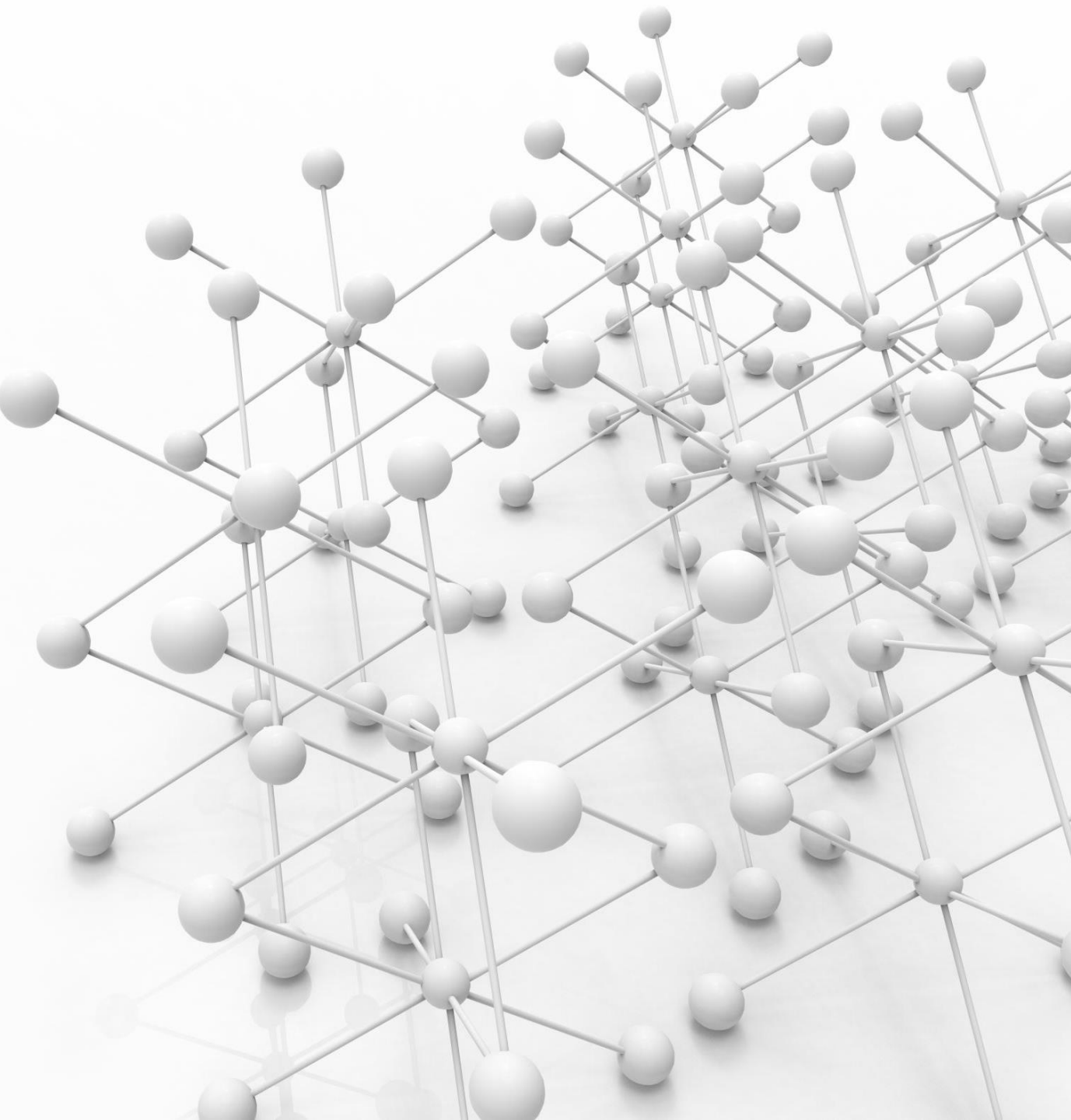




Data Structure

System Registration

Team's members

- 1-Momen Sherif Mohamed Ahmed 2401524751
Courses Linked List and Its functions and bonus feature
- 2- Marwan Mohammed Qassem Mohammed 2401249134
Enrolled Courses Linked List and Its functions
- 3-Youssef Sayed Hussien Saad 2401246144
Enrolled Students Linked List and Its functions
- 4-Mustafa Ahmed Ibrahim Ragab 2401245992
Students Linked List and Its Functions

=== University Course Registration System ===

1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

Enter your choice (1-17): 1

Enter student ID: 1

Student "1" Is Added Successfully

Adding a Student

```
=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 3

Enter course ID: 2001
The Course With ID: 2001 Is Added Successfully.
```

Adding a Course

```
=== University Course Registration System ===
```

1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

```
Enter your choice (1-17): 2
```

```
Enter student ID to remove: 1
```

```
Student "1" Is Deleted Successfully
```

Removing a Student

```
=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 4

Enter course ID to remove: 2001
The Course With ID: 2001 Is Deleted Successfully.
```

Removing a Course

```
=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 5

Enter student ID: 2
Enter course ID: 2002
Student 2 Enrolled In Course 2002 Successfully
Course enrolled successfully
```

Enrolling a Student

```
=== University Course Registration System ===
```

1. Add Student
2. Remove Student
3. Add Course
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

```
Enter your choice (1-17): 6
```

```
Enter student ID: 2
```

```
Enter course ID: 2002
```

```
Student 2 Removed From Course: 2002
```

```
Course: 2002 is empty now.
```

```
Enrollment Removed Successfully.
```

Unenrolling a Student


```
=== University Course Registration System ===
```

1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

```
Enter your choice (1-17): 7
```

```
List of All Students:
```

```
The Students List is:
```

```
4  
3  
2
```

Listing All Students

```
=== University Course Registration System ===
```

1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

```
Enter your choice (1-17): 8
```

```
List of All Courses:
```

```
the ID is: 2004
```

```
the ID is: 2003
```

```
the ID is: 2002
```

Listing All Courses

```
=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 9

Enter student ID: 2

Courses enrolled by student 2:
Course ID : 2004
Course ID : 2003
```

Listing Courses by Student

```
=== University Course Registration System ===
```

1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

```
Enter your choice (1-17): 10
```

```
Enter course ID: 2003
```

```
Students enrolled in course 2003:
```

```
Students Enrolled In Course: 2003
```

```
2
```

```
-----
```

Listing Students by Course

Enter your choice (1-17): 1

Enter student ID: 5

Student "5" Is Added Successfully

=== University Course Registration System ===

1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit

Enter your choice (1-17): 15

The Last Student Added Is "5"

Getting Last Student Added

```
Enter your choice (1-17): 3

Enter course ID: 2005
The Course With ID: 2005 Is Added Successfully.

=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 16

The Last Course Added Is "2005"
```

Getting Last Course Added

```
Enter your choice (1-17): 9

Enter student ID: 2

Courses enrolled by student 2:
Course ID : 2004
Course ID : 2003
Course ID : 2002

=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 13

Enter student ID to check: 2
Normal Student
```

Checking if a Student is Normal

```
Enter your choice (1-17): 10

Enter course ID: 2002
Students Enrolled In Course: 2002
3
2
-----

=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 14

Enter course ID to check: 2002
The Course Isn't Complete.
```

Checking if a Course is Complete


```
Enter student ID: 5
Student "5" Is Added Successfully

=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 11

Student "5" Is Deleted Successfully
Undo Done Successfully
```

Undoing Last Action

```
Enter your choice (1-17): 11

Student "5" Is Deleted Successfully
Undo Done Successfully

=== University Course Registration System ===
1. Add Student
2. Remove Student
3. Add Course
4. Remove Course
5. Enroll Student in Course
6. Remove Student from Course
7. List All Students
8. List All Courses
9. List Courses by Student
10. List Students by Course
11. Undo Last Action
12. Redo Last Action
13. Check if Student is Normal
14. Check if Course is Full
15. View Last Added Student
16. View Last Added Course
17. Exit
Enter your choice (1-17): 12

Student "5" Is Added Successfully
Redo Done Successfully
```

Redoing Last Action

Good bye